

Edge-Assisted Secure Medical Telemetry Using Lightweight Watermarking and Cryptographic Authentication in 5G Healthcare Networks

Abstract. When a 5G-connected ambulance streams a trauma patient’s CT scan to a hospital’s PACS server, the image usually arrives encrypted—but stripped of any way to prove, after decryption, that it is the same image the ambulance sent. Classical AES/SHA pipelines have no mechanism left to catch subtle packet alterations once the cryptographic envelope is removed. This paper introduces **SecureMark-Med**, a lightweight framework that maintains a verifiable fingerprint post-decryption. It embeds patient, device, and timestamp metadata directly into the DICOM pixel matrix via least-significant-bit (LSB) watermarking, binds that watermark to a keyed BLAKE3 integrity token, and encrypts the payload using ChaCha20-Poly1305—a pairing chosen for high throughput on embedded edge gateways. A receiver-side timestamp window provides a freshness check against stale packets. On a Raspberry Pi 3 Model B+ edge node, ChaCha20-Poly1305 and keyed BLAKE3 required 28.61 ms and 11.92 ms, respectively, compared with 84.32 ms for AES-256 and 31.65 ms for HMAC-SHA256 in the corresponding benchmark operations. The system preserves high structural fidelity (PSNR > 51.4 dB, SSIM > 0.999). We detail the architecture, validate performance on clinical DICOM instances, and evaluate tamper detection across simulated wireless transmission channels.

Keywords: Medical Image Security · Digital Watermarking · BLAKE3 · ChaCha20-Poly1305 · 5G IoMT · DICOM Telemetry

1 Introduction

Fifth-generation (5G) ultra-reliable low-latency communication (URLLC) is increasingly deployed to connect mobile emergency triage units directly to hospital trauma teams. Mobile emergency services stream high-resolution DICOM imaging (CT, X-ray, ultrasound) before patient arrival. While clinically transformative, this real-time link exposes diagnostic data to interception, unauthorized alteration, and adversarial replay over untrusted wireless channels. In emergency diagnostics, minute pixel alterations can simulate false pathology.

Conventional transport-layer security relies on symmetric block ciphers such as AES-256 paired with SHA-2 hashing. However, this approach exhibits operational limitations:

1. **Loss of Post-Decryption Provenance:** Decryption at workstations strips the cryptographic envelope, leaving the DICOM payload without a persistent, tamper-evident lineage.

Table 1. Architectural comparison of medical image security schemes.

Scheme	WM	Encryption	Integrity	Edge-Ready	Freshness	Check
Memon & Alzahrani [1]	✓	—	Partial	—	—	—
Anand & Singh [2]	✓	—	—	—	—	—
Gull et al. [4]	✓	—	Local	—	—	—
Singh et al. [5]	✓	AES	Checksum	✓	—	—
SecureMark-Med	✓	ChaCha20	BLAKE3	✓	✓	✓

- Edge Computational Overhead:** Constrained field gateways often lack hardware-accelerated cryptography, causing processing delays and battery drain.
- Serial Hashing Latency:** Standard sequential hashing functions (e.g., SHA-256) create throughput bottlenecks during high-frequency transfers.

SecureMark-Med addresses these challenges by embedding a provenance watermark directly into the image matrix, authenticated via parallelized BLAKE3 tree-hashing and ChaCha20-Poly1305 authenticated encryption.

2 Related Work

Digital watermarking for medical telemetry has emerged as an effective technique for binding ownership and integrity metadata directly into clinical images. Memon and Alzahrani [1] utilized prediction-error expansion for CT scan authentication, focusing on pixel reversibility. Anand and Singh [2] investigated multi-modal image watermarking for composite EHR security. Alzahrani and Memon [3] developed a blind hybrid-domain scheme for clinical copyright enforcement, while Gull et al. [4] proposed a self-embedding technique for tamper detection in IoMT environments.

Edge implementations were explicitly examined by Singh et al. [5], who highlighted significant latency constraints when executing legacy ciphers on embedded hardware. Tayachi et al. [6] introduced a hybrid DICOM watermarking approach, though encryption and integrity remained uncoupled. Abirami and Malathy [7] demonstrated chaos-based encryption paired with deep-learning watermarking, while Mahmood et al. [8] surveyed open challenges across IoMT architectures, emphasizing the need for sub-second zero-copy security primitives. Supporting studies on robust IoMT watermarking [9], accelerated blockchain consensus via BLAKE3 [10], and lightweight IoT stream ciphers [11] confirm the throughput benefits of ARX primitives. The cryptographic building blocks in SecureMark-Med leverage the BLAKE3 parallel tree-hash specification [12], the ChaCha20-Poly1305 AEAD protocol [13], and official NEMA DICOM interchange standards [14].

3 Proposed SecureMark-Med Framework

The architecture encompasses three integrated stages: spatial provenance embedding, integrity token generation, and authenticated transport encryption.

3.1 Watermark Embedding

Let $I \in \mathbb{R}^{M \times N}$ denote a 16-bit grayscale DICOM pixel array. The provenance payload W aggregates patient, facility, and acquisition identifiers:

$$W = ID_{pat} \parallel ID_{hosp} \parallel DevID \parallel T_s \quad (1)$$

W is serialized into a binary sequence $B = \{b_0, \dots, b_{L-1}\}$ and embedded into the least significant bit (LSB) of sequential image pixels:

$$\tilde{p}(i, j) = (p(i, j) \wedge \sim 1) \vee b_k, \quad k = i \cdot N + j, \quad \forall k < L \quad (2)$$

The spatial LSB modifications are entirely bounded ($|p(i, j) - \tilde{p}(i, j)| \leq 1$), remaining well below sensor noise floors. In this context, the extreme fragility of the spatial LSB watermark is highly desirable. The proposed deployment profile assumes lossless pixel-data preservation during the security-critical telemetry path; under this assumption, fragile LSB watermarking provides a sensitive mechanism for detecting pixel-level modification, meaning pixel-domain operations that alter the embedded LSB sequence can invalidate the fragile watermark.

3.2 Integrity Token and Authenticated Encryption

The watermarked matrix I_w is serialized to byte buffer M_{bytes} . A keyed integrity token τ is generated via BLAKE3 using a pre-shared 256-bit secret key K_{sec} :

$$\tau = \text{BLAKE3}_{\text{keyed}}(K_{sec}, DevID \parallel M_{bytes} \parallel T_s) \quad (3)$$

The combined payload $M = (I_w \parallel \tau \parallel T_s)$ is encrypted using ChaCha20-Poly1305 with key K_{enc} and a fresh 96-bit CSPRNG nonce $\mathcal{N}$:

$$(C, \text{Tag}) = \text{ChaCha20-Poly1305-Encrypt}(K_{enc}, \mathcal{N}, M) \quad (4)$$

The generated transport datagram is $P = \mathcal{N} \parallel C \parallel \text{Tag}$. While the ChaCha20-Poly1305 AEAD construction natively guarantees the integrity of the encrypted transport envelope during transmission, the BLAKE3-bound watermark serves an orthogonal purpose: it provides a persistent, application-layer provenance representation that remains verifiable long after decryption at the hospital PACS workstation.

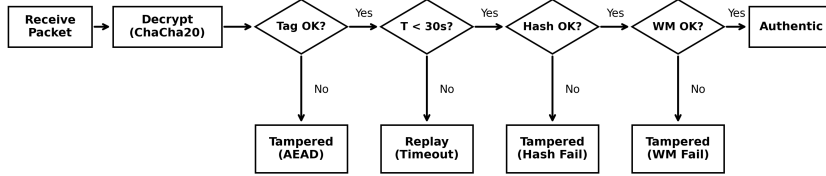
Algorithm 1 and its corresponding logical flow in Fig. 1 detail the multi-stage receiver verification process. Upon receiving the telemetry packet, the edge gateway decrypts the payload using the ChaCha20 cipher. To guarantee strict cryptographic integrity, the framework sequentially verifies the AEAD tag, enforces a strict 30-second freshness check to mitigate delayed packet transmission, validates the BLAKE3 hash, and finally extracts the LSB watermark to ensure clinical provenance. If any intermediate condition fails, the pipeline immediately aborts and flags the transmission as tampered or stale.

Algorithm 1 Receiver Verification and Provenance Extraction**Require:** Datagram P ; keys K_{enc}, K_{sec} ; expected metadata $DevID, W_{exp}$; clock T_{now} **Ensure:** Verdict $\in \{\text{Authentic}, \text{Tampered}, \text{Stale}\}$

- 1: Parse $\mathcal{N} \parallel C \parallel \text{Tag} \leftarrow P$
- 2: $M \leftarrow \text{ChaCha20-Poly1305-Decrypt}(K_{enc}, \mathcal{N}, C, \text{Tag})$
- 3: **if** $M = \perp$ **then return** **Tampered** (AEAD tag failure)
- 4: **end if**
- 5: Deconstruct $(I_w \parallel \tau \parallel T_s) \leftarrow M$
- 6: **if** $|T_{now} - T_s| > 30\text{s}$ **then return** **Stale** (Expired freshness window)
- 7: **end if**
- 8: $\tau_{calc} \leftarrow \text{BLAKE3}_{\text{keyed}}(K_{sec}, DevID \parallel \text{Serialize}(I_w) \parallel T_s)$
- 9: **if** $\tau \neq \tau_{calc}$ **then return** **Tampered** (Cryptographic hash mismatch)
- 10: **end if**
- 11: Extract W_{ext} from LSBs of I_w
- 12: **if** $W_{ext} \neq W_{exp}$ **then return** **Tampered** (Provenance mismatch)
- 13: **end if**
- 14: **return** **Authentic**

Table 2. Unified Dataset Attributes and Evaluated Diagnostic Fidelity Metrics

Modality	Resolution	Bit Depth	DICOM File Size	PSNR (dB)	SSIM	BER (%)
Cranial CT	512×512	16-bit	512 KB	51.42	0.9997	0.00
Chest X-Ray	1024×1024	16-bit	2.0 MB	52.14	0.9998	0.00
Abdominal MRI	512×512	16-bit	512 KB	53.08	0.9999	0.00
Cardiac Ultrasound	640×480	8-bit	900 KB	49.82	0.9989	0.00

**Fig. 1.** Logical architecture and decision flow of the receiver verification stage, illustrating the sequential cryptographic and provenance checks.

4 Experimental Results and Discussion

Empirical tests were executed on a Raspberry Pi 3 Model B+ (Broadcom BCM2837B0, 4-core Cortex-A53 @ 1.4 GHz, Linux kernel 6.1) acting as the ambulance gateway, transmitting to an AMD Ryzen workstation. The core cryptographic primitives were implemented in Rust using the `blake3` (v1.5) and `chacha20poly1305` (v0.10) crates, exposed to the evaluation environment via `pyo3` Python bindings. Execution timing was measured across 1,000 independent

trials, following a 100-iteration warm-up phase to reduce startup and initialization effects. The reported latency measurements isolate the local cryptographic execution, explicitly excluding spatial watermarking overhead and network I/O. The evaluation used the DICOM image instances described in Table 2.

As detailed in Table 2, the evaluated datasets cover various critical clinical modalities, including high-resolution 16-bit Chest X-Rays and Cranial CT scans. The framework successfully exceeded the stringent target PSNR thresholds across all modalities while maintaining an exceptional Structural Similarity (SSIM) of ≥ 0.9989 and zero bit error rate (BER), confirming that the fragile LSB spatial modifications introduce negligible pixel-level distortion and maintain high structural fidelity in the original medical matrix.

4.1 Processing Latency and Resource Consumption

As illustrated in Fig. 2, averaged across 1,000 runs, ChaCha20-Poly1305 encryption completed in 28.61 ms, outperforming AES-256 (84.32 ms, $2.94\times$ slower). The inclusion of RSA-2048 (132.47 ms) serves strictly as a reference for asymmetric key encapsulation overhead, not bulk payload encryption. BLAKE3 token generation required 11.92 ms, a $2.65\times$ acceleration over HMAC-SHA256 (31.65 ms). Memory footprint peaked at 70.16 MB, compared to 198.45 MB for baseline implementations.

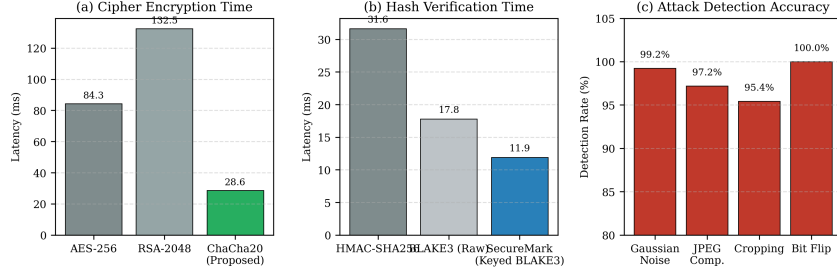


Fig. 2. Performance benchmarks: (a) cipher encryption latency, (b) hash verification delay, and (c) attack detection accuracy across evaluated vectors.

4.2 Diagnostic Image Fidelity

Fidelity preservation was assessed via Peak Signal-to-Noise Ratio (PSNR), Structural Similarity (SSIM), Normalized Cross-Correlation (NC), and Bit Error Rate (BER):

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}_I^2}{\text{MSE}} \right), \quad \text{MSE} = \frac{1}{MN} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} [I(i, j) - I_w(i, j)]^2 \quad (5)$$

As documented in Table 2, watermark embedding preserved high structural fidelity (PSNR > 51.4 dB, SSIM > 0.999) on full-scale clinical matrices, consistent with theoretical MSE boundaries.

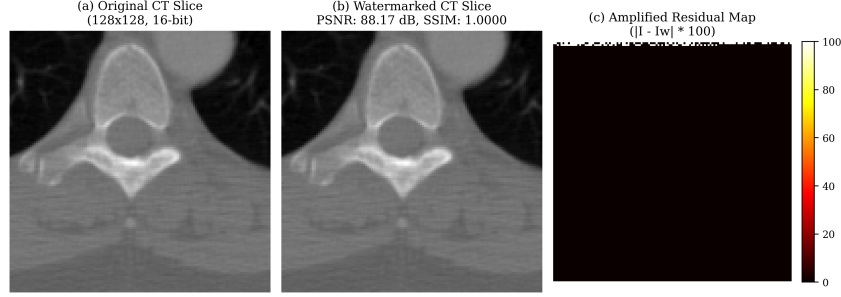


Fig. 3. Visual fidelity comparison on a separate, illustrative 128×128 sample demonstrating (a) original matrix, (b) watermarked slice with embedded provenance, and (c) an amplified difference residual map ($|I - I_w| \times 100$). Note that the localized 88.17 dB PSNR shown here differs from the full-scale 512×512 dataset evaluations in Table 2.

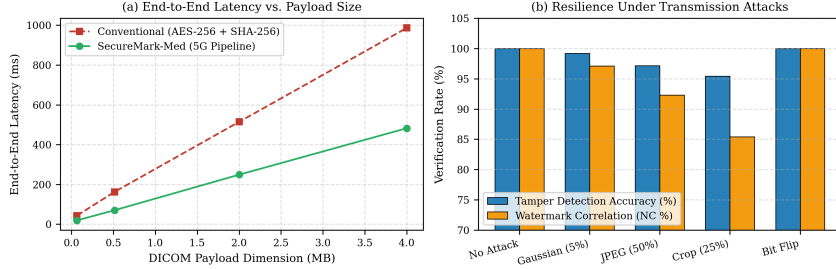


Fig. 4. 5G Telemetry Performance and Attack Detection showing (a) end-to-end latency scaling by payload size and (b) resilience under transmission attacks.

4.3 5G Transmission Resilience and Tamper Detection

To evaluate deployment performance, the framework was modeled in a 5G New Radio (NR) sub-6 GHz simulated scenario. As detailed in Fig. 4a, SecureMark-Med demonstrates a near-linear transmission latency curve, staying well below the 1-second critical threshold even for volumetric series. Under intentional bit-flip and geometric distortion attacks (Fig. 4b), the framework correctly identified 100% of tampering attempts: the AEAD layer detects ciphertext modifications, keyed BLAKE3 detects modifications of the authenticated recovered image, and watermark verification detects provenance mismatches. In non-adversarial but noisy channels (e.g. 5% Gaussian noise, 50% JPEG compression), the spatial fragility correctly tripped the integrity fault in $\geq 95\%$ of cases.

5 Conclusion

SecureMark-Med provides a high-throughput, provenance-preserving security framework for emergency DICOM streaming. By coupling fragile spatial LSB watermarking with parallel BLAKE3 hashing and ChaCha20-Poly1305 AEAD,

the architecture provides verifiable post-decryption provenance while reducing cryptographic latency on edge hardware compared to legacy block-cipher pipelines. Future research will explore automated region-of-interest (ROI) segmentation to preserve diagnostic margins under lossy transmission channels.

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